



## Bargaining over incentive contracts

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### ABSTRACT

The incentive contract theory assumes that the principal holds all of the bargaining power. By introducing alternating offers and strategic delay into the nonlinear pricing model, we relax this assumption and analyze an infinite-horizon contract bargaining game. We attain either the “sequential separating equilibrium” or the “simultaneous separating equilibrium”, depending on the parameter values. We prove the existence and the uniqueness of the equilibrium, and claim that multidimensionality and strategic delay can help resolve the multiple equilibria problem of bargaining theory. When the time between offers approaches zero, either quantity distortion or delay persists.

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### 1. Introduction

In many real-life situations, contracting involves bargaining. However, in the standard incentive contract theory the principal is assumed to hold all of the bargaining power. She proposes and commits to contracts in the form of a take-it-or-leave-it offer and therefore requests a yes-or-no answer from the agent. The agent is not free to propose another one (Salanie, 1997; Laffont and Martimort, 2002; Bolton and Dewatripont, 2005). If the agent rejects the offer, the interaction between them stops in the standard model, whereas in the real world, it actually might continue.

The assumption that the principal holds all of the bargaining power is reasonable if there is only one principal and many agents who compete for the principal's offer. For instance, in the case of employment contracts, potential workers compete for one job offered by the firm. However, in many real-life contracting situations, there is only one principal and one agent involved, and thus each party may have some bargaining power. For example, the firm and the labor union both have bargaining power when they negotiate the labor contract. In the bilateral monopoly situation, that is, one principal vs. one agent, it may be reasonable to assume that the agent also enjoys some bargaining power.

In this paper, we relax the standard assumption by introducing bargaining into the incentive contract theory. More precisely, we implement alternating offers (Rubinstein, 1982) to the standard non-linear pricing model (Mussa and Rosen, 1978; Maskin and Riley, 1984). We assume that after rejecting the principal's offer,

the agent can propose new contracts. In turn, the principal can choose to accept or reject those new contracts. By introducing alternating offers, we transform the standard one-shot contract design game into an infinite-horizon contract bargaining game.

In addition to the alternating offers assumption, we also allow the players to strategically use the time between offers, following Admati and Perry (1987). In other words, each party, when it is his or her turn to move, can delay it as long as he or she wishes. The assumption of strategic delay captures the idea that endogenous time between offers can be an essential strategic variable in bargaining with incomplete information (Admati and Perry, 1987).

In our contract bargaining game, a seller (principal) and a buyer (agent) bargain over a sales contract which specifies the price and the quantity of some good.<sup>1</sup> The seller has no private information, while the buyer does. In particular the buyer has either a high or a low valuation for the quantity. The principal makes the first offer. Subsequently, the players make alternating offers with strategic delay until they reach an agreement. In rounds when the uninformed seller proposes the contracts, it is a screening problem. The seller can offer a menu of contracts to induce the buyer to self-select. In rounds when the informed buyer proposes the contracts, it is a signaling problem. The buyer can choose delay or quantity to signal his type if necessary. Depending on the parameter values, we distinguish the following three cases: free screening, free signaling, and costly signaling. In the special case

<sup>1</sup> To my knowledge, Sen (2000) is the first paper analyzing this kind of game within the multidimensional bargaining setting. In the literature review, I will discuss Sen (2000), and the difference between Sen (2000) and our paper, in detail.

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of free screening, the equilibrium outcome is the same as that of the complete information. In the other two cases, we obtain two different equilibria as follows.

If the initial probability of the low-type buyer is sufficiently small, there exists a unique equilibrium. In this equilibrium, the seller's first offer is a single contract, which the high-type buyer accepts without delay while the low-type buyer does not. Instead, the low-type buyer makes a counteroffer. In the case of free signaling, the counteroffer is the low-type's complete-information contract; in the case of costly signaling, it is the "least-cost-separating contract" (Riley, 1979), which involves either quantity distortion or delay. Since the two type buyers are separated sequentially in the equilibrium path, we call it the "sequential separating equilibrium".

Also, if the initial probability of the low-type buyer is sufficiently large, there exists a unique equilibrium. In this equilibrium, the seller offers a menu of self-selection contracts without delay. Both type buyers accept the menu without delay and each selects his preferred contract. We call this unique equilibrium the "simultaneous separating equilibrium" and the contracts involved the "bargaining-proof contracts". Furthermore, when time between offers approaches zero, either delay or quantity distortion persist in the above two equilibria.

In the standard incentive contract theory, the contracts obtained should satisfy both the participation constraints (PCs) and the incentive compatibility constraints (ICs). In our contract bargaining game, the contracts negotiated should also meet these two constraints. However, for the contract design game, the right-hand sides of PCs usually are exogenous outside options. But for our contract bargaining game, the right-hand sides of PCs are the endogenous type-dependent continuation payoffs generated by sequential bargaining. Finding these endogenous continuation payoffs is the key to analyzing this type of bargaining game.

The main problem for the standard bargaining theory with incomplete information is that there are too many equilibria, even after applying refinements.<sup>2</sup> This problem is one of the reasons that the incentive contract theory makes the "extreme" assumption that the principal has all of the bargaining power (Myerson, 1983; Salanie, 1997). However, after introducing bargaining into the incentive contract model, this article obtains the unique equilibrium. In the cases of free screening and free signaling, we obtain the uniqueness without using any further refinements on the perfect Bayesian equilibrium; in the case of costly signaling, we obtain uniqueness by implementing the standard refinement "intuitive criterion" (Cho and Kreps, 1987). The reason for the uniqueness is because in our multidimensional bargaining game, the players can combine multidimensionality, as well as delay, as sorting devices, either signaling or screening credibly. Therefore, the incomplete information game can quickly become the complete information one. This is impossible in the standard one-dimensional bargaining games.

As mentioned before, Sen (2000) is one of the first papers analyzing the contract bargaining game by allowing alternating offers. Both Sen (2000) and our paper characterize the equilibria of the game. In the case when the high-type wishes to mimic the low-type, Sen (2000) characterizes the set of equilibrium outcomes,<sup>3</sup> while our paper establishes the uniqueness of the equilibrium explicitly.

Another main difference between Sen (2000) and our paper is that we introduce strategic delay into the contract bargaining game. The reasons for doing so are the following. First, Sen (2000) assumes that the two buyer types are close enough to ensure positive trade for both buyer types in equilibrium.<sup>4</sup> But we allow all possible parameter values for the two types. Under certain parameter values, the low-type cannot afford to signal by quantity distortion; he can signal by strategic delay only.<sup>5</sup> Second, when the low-type buyer can signal by both strategic delay and quantity distortion, he will choose the one which gives him higher payoff while separating him from the high-type. By doing so, our model endogenizes the choice of signals, while in most signaling literature, the signaling device is exogenously given (Spence, 1973; Riley, 2001). Meanwhile, we can find out the "real" least-cost separating equilibrium by comparing the low-type's payoffs from using different signals. Third, when the time between offers approaches zero, quantity distortion is the only inefficiency that persists in Sen (2000), while in our model, either quantity distortion or delay can persist, depending on the parameter values.<sup>6</sup>

There are other differences between Sen (2000) and our paper. For example, Sen (2000) has a more restrictive parameter value for the discount factor.<sup>7</sup> As a consequence, Sen (2000) excludes the free signaling case which is in our model. We implement the refinement of the "intuitive criterion"; while Sen (2000) combines the refinement of perfect sequential equilibrium (Grossman and Perry, 1986) and "elimination of never weak best response" (Kohlberg and Mertens, 1986) to prune the equilibrium set.

Besides Sen (2000), there are several other papers that focus on incentive contract bargaining. Wang (1998) discusses a one-sided offer case, in which only the uninformed party offers a menu each period. He shows that in every period the uninformed party will propose the same take-it-or-leave-it menu as in the standard incentive contract theory. The unique equilibrium outcome is the informed party accepts the menu and selects the contract without delay. Inderst (2003) allows both sides to propose menus. He proves that for a subset of parameters the bargaining game has a unique equilibrium where efficient contracts are implemented in the first period. Inderst (2002) shifts the bargaining power between the principal and the agent continuously, and concludes that endowing the informed party with more bargaining power reduces distortions.

Another closely related paper is Admati and Perry (1987), who analyze a bargaining model with strategic delay. Our model follows the bargaining rule of Admati and Perry (1987), but there are differences. First, our model is multidimensional bargaining, while theirs belongs to the standard one-dimensional bargaining game. Second, other than strategic delay, our model has more signaling variables, such as quantity distortion. The player can sometimes endogenously choose which signal to use as mentioned above. Third, we obtain unique equilibrium, while Admati and Perry (1987) still have multiple equilibria when the parameter values are in the intermediate ranges, even after using the "intuitive criterion". Cramton (1992) extends Admati and Perry (1987) to two-sided uncertainty with continuous type settings.

The basic structure of our model is developed in Section 2. Section 3 characterizes the equilibrium and proves its existence and uniqueness. Section 4 concludes.

<sup>4</sup> Please see Assumption 2 and the related explanation for it in Sen (2000).

<sup>5</sup> See Section 3.2.1 for more details.

<sup>6</sup> As an application, for example, Sen (2000) cannot explain why strikes still happen given that the labor contracts are multidimensional in management vs. union bargaining. But our model has implications about when strikes will happen and when they will not.

<sup>7</sup> Please refer to Lemma 3 of Sen (2000).

<sup>2</sup> See Binmore et al. (1992), Kennan and Wilson (1993) and Ausubel et al. (2002) for surveys on bargaining theory with incomplete information.

<sup>3</sup> Please refer to Propositions 5 and 6 of Sen (2000) and the related explanations there. In Proposition 6, Sen (2000) does show a unique equilibrium outcome, but it requires that the time between offers approaches zero.

## 2. The model

A seller (principal) and a buyer (agent) bargain over quantity  $q \in \mathbb{R}_+$  and total price (payment)  $p \in \mathbb{R}_+$  of some good. The seller's cost of quantity  $q$  is  $C(q)$ , and  $C(\cdot)$  is twice continuously differentiable with  $C' > 0$  and  $C'' > 0$ . The seller has no private information and her payoff function is  $V(q, p) = p - C(q)$ . But the buyer has private information about his own valuation, which is either high or low and denoted by  $\theta \in \{\underline{\theta}, \bar{\theta}\}$ , with  $0 < \underline{\theta} < \bar{\theta}$ . The payoff function for the buyer is  $U(q, p, \theta) = R(q, \theta) - p$ , where  $R(q, \theta)$  is twice continuously differentiable with respect to  $q$ ,  $R(q, \bar{\theta}) > R(q, \underline{\theta})$  for all  $q$ , and  $R(q, \bar{\theta}) - R(q^0, \bar{\theta}) > R(q, \underline{\theta}) - R(q^0, \underline{\theta})$  for any pair  $q > q^0$ . The seller believes that the buyer is type  $\underline{\theta}$  with probability  $\pi$ , and this prior belief is common knowledge. Denote the total surplus  $S(q, \theta) = R(q, \theta) - C(q)$ , which is strictly concave and maximized by a finite value  $q^*(\theta)$ .  $S(q^*(\theta), \theta)$  is the maximum amount of surplus attainable. Denote  $\bar{S}^* \equiv S(q^*(\bar{\theta}), \bar{\theta})$  and  $\underline{S}^* \equiv S(q^*(\underline{\theta}), \underline{\theta})$ , and we have  $\bar{S}^* > \underline{S}^* > 0$ .

### 2.1. The contract bargaining game

Following Admati and Perry (1987), the contract bargaining game starts at time zero, and the seller makes the first offer.<sup>8</sup> The seller can propose a menu of contracts  $(\underline{m}, \bar{m})$ , where  $\underline{m} = (q, p)$ ,  $\bar{m} = (\bar{q}, \bar{p})$ , from which, after acceptance, the buyer is free to choose one contract from the menu.<sup>9</sup> After the rejection of the entire menu, the buyer can offer a contract.<sup>10</sup> If the seller rejects the contract, she again proposes a menu, and so on so forth. A response to an offer involves either an acceptance or a rejection plus a counteroffer, and it is made within no shorter than a given length of time  $\Delta$ , which is normalized to one. However, the response can be made at any time later. Acceptance ends the game. We assume that until a counteroffer is made, the other party cannot change his or her previous offer.<sup>11</sup> The first offer in the game can be made at any time  $t \geq 0$ .

We use  $\tau \geq 0$  to denote strategic delay, i.e., the length of the time beyond the minimal time of one it takes the relevant player to make a response to an existing offer. We say “without delay” when  $\tau = 0$ .

A relevant history for the game is a sequence of unacceptable offers and a sequence of time lengths between offers. Formally, if  $n \in \{1, 2, \dots, \infty\}$  is the number of rounds (offers), then a history of the  $n$  rounds is denoted by  $h^n = \{m^j, \tau^j\}_{j=1}^n$ . Note that  $\tau^n$  denotes the time delay above the minimum time of one in round  $n$  since offer  $m^{n-1}$  was made, and this history indicates that  $t = n + \sum_{j=1}^n \tau^j$  units of time have passed. A strategy for a player specifies for each history after which it is the player's turn to move, the length of time delay  $\tau$ , whether the latest offer is accepted, and if not accepted, a counteroffer. Let  $\pi(h^n)$  denote the seller's belief at  $h^n$  that the buyer is a low-type. We consider only pure strategies.

The equilibrium concept that we use is the perfect Bayesian equilibrium (PBE), which requires that the strategies are optimal given the beliefs, and the beliefs are obtained from equilibrium strategies and observed actions using Bayes' rule whenever

possible.<sup>12</sup> An outcome of the game is  $(m(q, p), t)$ , with the interpretation that, contract  $m$  is agreed on at time  $t$ , the buyer pays  $p$  to the seller to obtain quantity  $q$ . Let  $\delta$  be the common discount factor. The seller's payoff is  $\delta^{t-1}V(m)$ , while the buyer's payoff is either  $\delta^{t-1}\underline{U}(m)$  for the low-type or  $\delta^{t-1}\bar{U}(m)$  for the high-type.

### 2.2. The complete information outcome

As a benchmark, first consider the complete information outcome. Following Rubinstein (1982), we can obtain the complete information contracts with alternating offers made by the seller and the buyer respectively:

$$\bar{m}_s^* = \left( \bar{q}^*, C(\bar{q}^*) + \frac{1}{1+\delta}\bar{S}^* \right) \quad (1)$$

$$\underline{m}_s^* = \left( \underline{q}^*, C(\underline{q}^*) + \frac{1}{1+\delta}\underline{S}^* \right) \quad (2)$$

$$\bar{m}_b^* = \left( \bar{q}^*, C(\bar{q}^*) + \frac{\delta}{1+\delta}\bar{S}^* \right) \quad (3)$$

$$\underline{m}_b^* = \left( \underline{q}^*, C(\underline{q}^*) + \frac{\delta}{1+\delta}\underline{S}^* \right) \quad (4)$$

(1) and (2) are the contracts that the seller provides to the high-type and the low-type buyer, respectively. (3) and (4) are the contracts for the seller proposed by the high-type and the low-type buyer, respectively. Since this is complete information bargaining, all of the quantities in these contracts are first-best and the payments divide the total surplus between the seller and the buyer.

**Remark 1** (Rubinstein, 1982). When the seller has complete information about the buyer, there is a unique subgame perfect equilibrium. In this equilibrium, the seller offers  $\bar{m}_s^*$  ( $\underline{m}_s^*$ ) to the high-type (low-type) buyer whenever it is her turn to make an offer, and accepts an offer  $\bar{m}_b$  ( $\underline{m}_b$ ) of the buyer with type  $\bar{\theta}$  ( $\underline{\theta}$ ) if and only if  $V(\bar{m}_b) \geq \frac{\delta}{1+\delta}\bar{S}^*$  ( $V(\underline{m}_b) \geq \frac{\delta}{1+\delta}\underline{S}^*$ ); the buyer with type  $\bar{\theta}$  ( $\underline{\theta}$ ) always proposes  $\bar{m}_b^*$  ( $\underline{m}_b^*$ ), and accepts an offer  $\bar{m}_s$  ( $\underline{m}_s$ ) if and only if  $\bar{U}(\bar{m}_s) \geq \frac{\delta}{1+\delta}\bar{S}^*$  ( $\underline{U}(\underline{m}_s) \geq \frac{\delta}{1+\delta}\underline{S}^*$ ). The outcome is that the seller proposes  $\bar{m}_s^*$  ( $\underline{m}_s^*$ ) at time zero, and the buyer with type  $\bar{\theta}$  ( $\underline{\theta}$ ) accepts this offer without delay.

In the above equilibrium, the seller gets  $\frac{1}{1+\delta}$  share of total surplus  $\bar{S}^*$  ( $\underline{S}^*$ ), and the buyer with type  $\bar{\theta}$  ( $\underline{\theta}$ ) gets  $\frac{\delta}{1+\delta}$  share of total surplus  $\bar{S}^*$  ( $\underline{S}^*$ ).

## 3. Screening vs. signaling: free or costly

Now we go back to the incomplete information scenario. In rounds when the uninformed party proposes the contracts, it is a screening problem; in rounds when the informed party makes an offer, it is a signaling problem. In this section, we first review the case of free screening, then we analyze the case of free signaling, and finally we discuss the case of costly signaling.

It is simple to show that the low-type buyer will never mimic the high-type's complete information strategy because  $\underline{U}(m_s^*) > \underline{U}(\bar{m}_s^*)$  and  $\underline{U}(m_b^*) > \underline{U}(\bar{m}_b^*)$ . Then the question is whether the high-type will mimic the low-type's complete information strategy or not.

<sup>8</sup> Admati and Perry (1987) implicitly assume that the first offer takes no time. See also Cramton (1992) for a modification. But we still follow Admati and Perry (1987).

<sup>9</sup> Since we allow only two buyer types, without loss of generality, we can assume the menu offered consists of two contracts. The contracts in the menu could be identical.

<sup>10</sup> One reason for this assumption is because the seller has no private information. Sen (2000) discusses and justifies this assumption.

<sup>11</sup> Admati and Perry (1987) discuss the assumption, and Cramton (1992) provides a justification for this assumption.

<sup>12</sup> For details about the PBE, please see Fudenberg and Tirole (1991).

We first review a special case: free screening. Contrary to the standard costly screening problem, free screening means that the seller can distinguish the different buyer types with the complete information contracts defined above, despite the incomplete information scenario. Therefore, the first-best efficient outcome can be obtained. The following is the result:

**Remark 2** (Inderst, 2003). When  $\delta$  is sufficiently large such that the following condition holds:

$$\delta(\bar{S}^* - \underline{S}^*) \geq (1 + \delta)(R(\underline{q}^*, \bar{\theta}) - R(\underline{q}^*, \underline{\theta})), \quad (C1)$$

the contract bargaining game has a unique equilibrium outcome where the seller proposes  $(\bar{m}_s^*, \underline{m}_s^*)$  at time zero, and the high-type (low-type) buyer accepts this offer without delay and selects  $\bar{m}_s^*$  ( $\underline{m}_s^*$ ).<sup>13</sup>

Different from the usual contract design game, the alternating offer bargaining creates endogenous type-dependent continuation payoffs. Under certain conditions, the incentive compatibility constraints are satisfied by the complete information contracts. One can easily verify that (C1) is satisfied if and only if  $\bar{U}(\bar{m}_s^*) \geq \bar{U}(\underline{m}_s^*)$ . Since  $\bar{U}(\bar{m}_b^*) - \bar{U}(\underline{m}_b^*) = \bar{U}(\bar{m}_s^*) - \bar{U}(\underline{m}_s^*) + \frac{1-\delta}{1+\delta}(\bar{S}^* - \underline{S}^*)$ ,  $\bar{U}(\bar{m}_s^*) \geq \bar{U}(\underline{m}_s^*)$  implies  $\bar{U}(\bar{m}_b^*) > \bar{U}(\underline{m}_b^*)$ . Hence, under condition (C1), the high-type buyer has no incentive to mimic the low-type buyer's complete information strategy. As mentioned above, the low-type will never mimic the high-type. Therefore, the equilibrium and the outcome are identical to those of the complete information, and the first-best is achieved despite the incomplete information.

### 3.1. Free signaling

From now on, we focus on the case in which (C1) is violated or  $\bar{U}(\bar{m}_s^*) < \bar{U}(\underline{m}_s^*)$ . Therefore, free screening is impossible. But we can discuss another related case: free signaling, which means that the low-type buyer can signal his private information by his complete information contract when it is his turn to make an offer.

Consider parameters such that for  $0 < \delta < 1$ , the following condition holds:

$$\delta(\bar{S}^* - \underline{S}^*) < (1 + \delta)(R(\underline{q}^*, \bar{\theta}) - R(\underline{q}^*, \underline{\theta})) \leq \bar{S}^* - \underline{S}^*. \quad (C2)$$

It is easy to verify that under condition (C2), we have  $\bar{U}(\bar{m}_s^*) < \bar{U}(\underline{m}_s^*)$  and  $\bar{U}(\bar{m}_b^*) \geq \bar{U}(\underline{m}_b^*)$ . In this situation, even though the seller cannot screen for free, the buyer can signal for free. In other words, the high-type can attain a higher continuation payoff by revealing his type than mimicking the low-type when it is the buyer's turn to make an offer. The intuition behind signaling for free is because the high-type faces a trade-off here. By faking to be the low-type, the high-type will pay less to the seller. However, the total surplus also shrinks since less quantities are traded. Under condition (C2), the high-type will obtain less by faking to be the low-type.

#### 3.1.1. The equilibrium

Assume that condition (C2) holds. We first obtain the following lemma:

**Lemma 1.** In the case of free signaling, when the buyer makes an offer, the unique payoff for the high-type buyer is  $\bar{U}(\bar{m}_b^*)$  or  $\frac{1}{1+\delta}\bar{S}^*$ ; the unique payoff for the low-type is  $\underline{U}(\underline{m}_b^*)$  or  $\frac{1}{1+\delta}\underline{S}^*$ .

In the case of free signaling, the buyer can signal his type by his complete information contract. Therefore, when it is the buyer's turn to make an offer, the situation is just like the case of complete information and both type buyers will offer their corresponding complete information contracts. That is, the high-type buyer will propose  $\bar{m}_b^*$  and the low-type will propose  $\underline{m}_b^*$ , and their payoffs are their corresponding complete information payoffs.

With respect to both type buyers' unique continuation payoffs, the seller's problem (SP) can be written as follows when it is her turn to make the offer:

$$(SP) \quad V = \max_{\{(\underline{p}, \underline{q}); (\bar{p}, \bar{q})\}} \{\pi[\underline{p} - C(\underline{q})] + (1 - \pi)[\bar{p} - C(\bar{q})]\}$$

$$\text{s.t. } R(\underline{q}, \underline{\theta}) - \underline{p} \geq R(\bar{q}, \underline{\theta}) - \bar{p} \quad (IC)$$

$$R(\bar{q}, \bar{\theta}) - \bar{p} \geq R(\underline{q}, \bar{\theta}) - \underline{p} \quad (\bar{IC})$$

$$R(\underline{q}, \underline{\theta}) - \underline{p} \geq \frac{\delta}{1 + \delta} \underline{S}^* \quad (PC)$$

$$R(\bar{q}, \bar{\theta}) - \bar{p} \geq \frac{\delta}{1 + \delta} \bar{S}^*. \quad (\bar{PC})$$

As usual, (IC) and ( $\bar{IC}$ ) are the incentive compatibility constraints, while (PC) and ( $\bar{PC}$ ) are the participation constraints for the low-type and the high-type, respectively. Note that the right-hand sides of the participation constraints are the continuation payoffs for the two type buyers, respectively.

Solving (SP), we find that there are two possible equilibria, and each of them is unique under certain parameter values as specified in Proposition 1.

The simultaneous separating equilibrium (SIE1). In this equilibrium, the seller proposes a menu consisting of two self-selection contracts at time zero, and each type buyer accepts this menu without delay and selects the contract for himself. The menu proposed by the seller is  $\{(q^0, p^0); (\bar{q}^*, \bar{p}^0)\}$ , where  $q^0$  is defined by the first-order condition  $C'(q) = R'(q, \underline{\theta}) - \frac{1-\pi}{\pi}[R'(q, \bar{\theta}) - R'(q, \underline{\theta})]$ ,<sup>14</sup> and  $q^0 < \underline{q}^*$ . Correspondingly,  $\underline{p}^0 = R(q^0, \underline{\theta}) - \frac{\delta}{1+\delta}\underline{S}^*$  and  $\bar{p}^0 = R(\bar{q}^*, \bar{\theta}) - R(q^0, \bar{\theta}) + \underline{p}^0$ . Under this equilibrium, the expected payoff for the seller is:

$$V_1 = \pi[\underline{p}^0 - C(q^0)] + (1 - \pi)[\bar{p}^0 - C(\bar{q}^*)].$$

The sequential separating equilibrium (SEE1). In this equilibrium, the seller proposes a single contract  $\bar{m}_s^*$  at time zero, and the high-type buyer accepts it without delay, but the low-type buyer rejects the offer and makes counteroffer  $\underline{m}_b^*$  without delay, which the seller accepts without delay. Under this equilibrium, the expected payoff for the seller is:

$$V_2 = \pi \frac{\delta^2}{1 + \delta} \underline{S}^* + (1 - \pi) \frac{1}{1 + \delta} \bar{S}^*.$$

Comparing  $V_1$  and  $V_2$ , we obtain the following proposition:

**Proposition 1.** Under condition (C2), there exists value  $0 < \pi^0 < 1$  such that when  $\pi \leq \pi^0$ , there exists the unique sequential separating equilibrium (SEE1); when  $\pi > \pi^0$ , there exists the unique simultaneous separating equilibrium (SIE1).

<sup>13</sup> Sen (2000) and Inderst (2003) get similar efficient results, and Inderst (2003) proves the uniqueness of the equilibrium.

<sup>14</sup> When solving (SP), it can be shown that ( $\bar{IC}$ ) and (PC) are binding. Replacing them into the objective function, we can obtain this first-order condition. For  $C'(q) > 0$ , it is necessary that  $\pi > 1 - \frac{R'(q, \underline{\theta})}{R'(q, \bar{\theta})}$ ; otherwise,  $q^0 = 0$ .

When  $\pi$  is sufficiently small, the seller will propose the contract only for the high-type buyer and exclude the low-type. When  $\pi$  is sufficiently large, the seller will propose a menu consisting of self-selection contracts. Given the seller's offer, the buyer will respond accordingly. We prove the existence of the equilibrium by first specifying the players' strategies and the seller's (out-of-equilibrium) beliefs. Then we show that the strategies are optimal given the beliefs, and the beliefs are updated by using Bayes' rule whenever possible. The proof of the existence is provided in the Appendix.

In Proposition 1, we also claim that the equilibrium specified above is unique. We prove the uniqueness by the following two lemmas.

**Lemma 2.** *In the free signaling case, when  $\pi > \pi^0$ ,  $V_1$  is the unique equilibrium payoff for the seller; when  $\pi \leq \pi^0$ ,  $V_2$  is the unique equilibrium payoff for the seller.*

**Proof.** First, the seller's problem (SP) is well-defined. It takes each type buyer's unique continuation payoff as his outside option in the corresponding participation constraint. Second, as we have shown above, when  $\pi > \pi^0$  the menu  $\{(q^0, p^0); (\bar{q}^*, \bar{p}^0)\}$  is the unique solution to (SP). As a consequence,  $\bar{V}_1$  is the unique payoff for the seller. When  $\pi \leq \pi^0$ ,  $\bar{m}_s^*$  is the unique solution to (SP), which is accepted by the high-type buyer only. After rejecting  $\bar{m}_s^*$ , the low-type proposes his complete information contract without delay and obtains his unique payoff. Thus,  $V_2$  is the unique payoff for the seller in this case, too.  $\square$

Both type buyers' continuation payoffs are unique and the seller's problem (SP) is well-defined, therefore, the solution to (SP) is unique. As a consequence, the seller's payoff is also unique.

**Lemma 3.** *In the free signaling case, when  $\pi > \pi^0$ , the only equilibrium path is SIE1; when  $\pi \leq \pi^0$ , the only equilibrium path is SEE1.*

**Proof.** When  $\pi > \pi^0$ , as we have shown above, menu  $\{(q^0, p^0); (\bar{q}^*, \bar{p}^0)\}$  is the unique solution to (SP). With this menu, the high-type buyer has a higher payoff than when he proposes, and the low-type gets his unique payoff as Lemma 1 shows. Both types will accept this offer without delay. In addition,  $V_1$  is the unique equilibrium payoff for the seller as Lemma 2 shows. Thus, the SIE1 is the only candidate for an equilibrium path. The existence of an equilibrium that supports this path has been established.

When  $\pi \leq \pi^0$ , it is the seller's best interest to offer a single contract  $\bar{m}_s^*$ . Meanwhile, in the equilibrium path of SEE1, both type buyers obtain their corresponding complete information payoffs. In addition,  $V_2$  is the unique equilibrium payoff for the seller as Lemma 2 shows. Therefore, SEE1 is the unique equilibrium path when  $\pi \leq \pi^0$ .  $\square$

Note that without any further refinements, in the above cases, the PBE for the contract bargaining game is unique, despite incomplete information. The reason is because bargaining with alternating offers generates type-dependent continuation payoffs. When (C1) or (C2) are satisfied, the players can either screen for free or signal for free, which simplifies the analysis and helps to obtain the uniqueness without any further refinements of the PBE.

### 3.2. Costly signaling

Now consider parameters such that  $\bar{U}(\bar{m}_s^*) < \bar{U}(\underline{m}_s^*)$  and  $\bar{U}(\bar{m}_b^*) < \bar{U}(\underline{m}_b^*)$ , or

$$(1 + \delta)(R(\bar{q}^*, \bar{\theta}) - R(\underline{q}^*, \underline{\theta})) > \bar{S}^* - \underline{S}^*. \quad (C3)$$

Under condition (C3), the high-type has the incentive to mimic the low-type. If he wishes to separate himself from the high-type, the low-type buyer has to signal his type costly. Before proceeding to the analysis, we make Assumption 1 in the spirit of the "intuitive criterion" (Cho and Kreps, 1987).

**Assumption 1.** (A1). Suppose that, according to the equilibrium strategies, either (i) at time  $t$ , the seller accepts an offer  $m$  made by the buyer, or (ii) at time  $t$ , the high-type accepts an offer  $m$  made by the seller. If the buyer offers  $\tilde{m}$  at time  $t + \tau$ , and  $(\tilde{m}, \tau)$  satisfies  $\underline{U}(\tilde{m}, t + \tau) > \underline{U}(m, t)$ ,  $\bar{U}(\tilde{m}, t + \tau) < \bar{U}(m, t)$  and  $V(\tilde{m}, t + \tau) \geq V(\underline{m}_b^*, t + \tau)$ , then the belief of the seller upon getting this offer must be  $\underline{\pi} = 1$ .

Assumption 1 says that if a deviation by the buyer is profitable for the low-type but not for the high-type, then the deviation should be attributed to the low-type.<sup>15</sup>

#### 3.2.1. Choice of signals

First consider a situation where the seller's current belief is her prior, and it is the buyer's turn to make an offer. The high-type wants to pool with the low-type. However, either the seller will reject the pooling offer, or the low-type has no incentive to pool with the high-type:

**Lemma 4.** *The pooling offer cannot exist in the equilibrium path.*

Note that if the pooling offer is the low-type's complete information contract  $\underline{m}_b^*$ , the seller will reject this offer given  $\pi < 1$ . Suppose that the pooling offer gives the seller more than  $\frac{\delta}{1+\delta}\bar{S}^*$  and is acceptable to the seller. Then the low-type buyer can profitably deviate to a separating offer, following the argument of the "intuitive criterion".

From now on, we will focus on the "least-cost separating equilibrium" (Riley outcome) because the "intuitive criterion" selects it in the two-type signaling model. The following two lemmas are useful for our analysis:

**Lemma 5.** *In the least-cost separating equilibrium, the high-type buyer will make offer  $\bar{m}_b^*$  without delay and obtain  $\frac{1}{1+\delta}\bar{S}^*$ .*

Since the low-type has separated himself from the high-type by costly signal in the least-cost separating equilibrium, there is no reason for the high-type to delay his offer, or to distort the quantity. Otherwise, he would receive strictly less than  $\frac{1}{1+\delta}\bar{S}^*$ .

**Lemma 6.** *If the seller's posterior belief is  $\underline{\pi} = 1$ , then the seller accepts any offer no less than  $V(\underline{m}_b^*)$  or  $\frac{\delta}{1+\delta}\underline{S}^*$ .*

When the seller believes the buyer is a low-type, the seller should obtain at least her complete information payoff when it is the seller's turn to respond to the low-type's offer. Otherwise, she can deviate profitably.

In our model, there are two alternative ways that the low-type may use to signal his type: quantity distortion or strategic delay. The low-type buyer will choose the signal which gives him higher payoff, while separating himself from the high-type. By doing so, we endogenize the choice of signals, while in most signaling literature the signaling device is exogenously given.<sup>16</sup>

<sup>15</sup> Note that when the deviation  $(\tilde{m}, \tau)$  happens, for the deviation to be meaningful, it also requires that  $V(\tilde{m}, t + \tau) \geq V(\underline{m}_b^*, t + \tau)$ . In other words, the meaningful deviation should satisfy the seller's participation constraint. Otherwise, no information is transmitted to the seller and the seller will ignore such kind of deviation.

<sup>16</sup> See Riley (2001) for a survey.

However, under certain circumstances, the signaling cost by quantity distortion could be so high that the low-type cannot afford to use it. This is possible because when the two buyer types are sufficiently far apart, the quantity distortion could be so significant that after giving the seller her complete information payoff  $\frac{\delta}{1+\delta}\bar{S}^*$ , the low-type's payoff would be negative.<sup>17</sup> Then the low-type will signal by delay only, which can guarantee him a positive payoff.

We first discuss the case involving only strategic delay as the signal. Then we discuss the case supposing that the low-type signals by quantity distortion. Finally, we compare the payoffs of these two cases to determine which signal the buyer will use.

Note that when the low-type signals his type only by strategic delay, he proposes the complete information contract  $\underline{m}_b^*$  with delay  $\tau$ , and faces the following problem:

$$\begin{aligned} \max_{(\tau)} \underline{U}(\tau) &= \delta^\tau (R(\underline{q}^*, \underline{\theta}) - \underline{p}_b^*) \\ \text{s.t. } \frac{1}{1+\delta}\bar{S}^* &\geq \delta^\tau (R(\underline{q}^*, \bar{\theta}) - \underline{p}_b^*), \end{aligned} \quad (\text{IC}^1)$$

where  $\underline{p}_b^* \equiv C(\underline{q}^*) + \frac{\delta}{1+\delta}\bar{S}^*$ . (IC<sup>1</sup>) is the incentive compatibility constraint for the high-type, which ensures he will not mimic the low-type's offer. With offer  $\underline{m}_b^*$ , the participation constraint for the seller is satisfied automatically. Solving this problem, we can obtain  $\tau^*$  directly since (IC<sup>1</sup>) is binding. Thus  $\underline{U}(\tau^*) = \delta^{\tau^*}(\frac{1}{1+\delta}\bar{S}^*)$ . We then have the following lemma:

**Lemma 7.** *When the low-type signals by strategic delay, there is a unique least-cost separating equilibrium: (1) The high-type buyer offers his complete information contract  $\bar{m}_b^*$ ; (2) The low-type buyer offers his complete information contract  $\underline{m}_b^*$  with delay  $\tau^*$ ; (3) The seller's beliefs are  $\pi(\bar{m}_b^*) = 0$  and  $\pi(\underline{m}_b^*, \tau^*) = 1$ ; (4) If a buyer offers a contract  $m$  with delay  $\tau$ , and  $(m, \tau)$  satisfies  $\bar{U}(m, \tau) > \bar{U}(\bar{m}_b^*)$ ,  $\underline{U}(m, \tau) < \underline{U}(\underline{m}_b^*, \tau^*)$ , then the belief of the seller is  $\pi' = 0$ .*

When the low-type buyer signals by the quantity distortion, he solves the following problem:

$$\begin{aligned} \max_{(q, p)} \underline{U}(q, p) &= R(q, \underline{\theta}) - p \\ \text{s.t. } p - C(q) &\geq \frac{\delta}{1+\delta}\bar{S}^* \\ \frac{1}{1+\delta}\bar{S}^* &\geq R(q, \bar{\theta}) - p \\ \underline{U}(q, p) &> 0 \end{aligned} \quad \begin{aligned} &(\text{PC}^2) \\ &(\text{IC}^2) \\ &(\text{PC}^2) \end{aligned}$$

(PC<sup>2</sup>) is the seller's participation constraint. (IC<sup>2</sup>) is the incentive compatibility constraint for the high-type. The left-hand side of (IC<sup>2</sup>) is due to Lemma 5. (PC<sup>2</sup>) ensures that the low-type's payoff is positive after signaling. Solving this problem, we can obtain the following result:

$$(q, p) = \begin{cases} (q_1, p_1), & \text{if } \underline{U}(q_1, p_1) > 0 \\ (0, 0), & \text{otherwise,} \end{cases}$$

where  $q_1$  is the smaller solution to  $R(q, \bar{\theta}) - C(q) = \frac{1}{1+\delta}\bar{S}^* + \frac{\delta}{1+\delta}\bar{S}^*$  with  $q_1 < q^*$ ,  $p_1 = C(q_1) + \frac{\delta}{1+\delta}\bar{S}^*$ , and  $\underline{U}(q_1, p_1) = R(q_1, \underline{\theta}) - C(q_1) - \frac{\delta}{1+\delta}\bar{S}^* > 0$ . In this case, (PC<sup>2</sup>) is satisfied. However, if (PC<sup>2</sup>) is violated, then  $(q, p) = (0, 0)$  and  $\underline{U}(0, 0) = 0$ . Therefore, the low-type cannot signal by quantity distortion. He can only use delay as the signal. We thus obtain the following lemma when  $\underline{U}(q_1, p_1) > 0$ :

**Lemma 8.** *When the low-type can signal by quantity distortion, there is a unique least-cost separating equilibrium: (1) The high-type buyer offers his complete-information contract  $\bar{m}_b^*$ ; (2) The low-type buyer offers the contract  $(\underline{p}_1, \underline{q}_1)$ ; (3) The seller's beliefs are  $\pi(\bar{m}_b^*) = 0$  and  $\pi(\underline{p}_1, \underline{q}_1) = 1$ ; (4) If a buyer offers a contract  $m$  with delay  $\tau$ , and  $(m, \tau)$  satisfies  $\bar{U}(m, \tau) > \bar{U}(\bar{m}_b^*)$ ,  $\underline{U}(m, \tau) < \underline{U}(\underline{p}_1, \underline{q}_1)$ , then the belief of the seller is  $\pi'' = 0$ .*

The low-type buyer will compare  $\underline{U}(q, p)$  and  $\underline{U}(\tau^*)$  to decide which signal to use:

**Proposition 2.** *If  $\underline{U}(q, p) > 0$  and the following condition holds:*

$$(1 + \delta)(R(\underline{q}_1, \underline{\theta}) - C(\underline{q}_1)) \geq (\delta + \delta^{\tau^*})\bar{S}^*, \quad (\text{C4})$$

*the low-type will signal by quantity distortion; however, when (C4) is violated or  $\underline{U}(q, p) = 0$ , he will signal by delay.*

As we have shown above, in the quantity distortion case, the low-type has to bear all of the signaling costs; in the strategic delay case, he shares the signaling costs with the seller, but delay can be more costly. Proposition 2 says that when the low-type can signal by both the quantity distortion and the delay, he will compare the payoffs between them and choose the signal which gives him the higher payoff. However, if the low-type cannot signal by the quantity distortion, he will use the delay signal only.

Define  $\underline{U}_0 \equiv \max\{\underline{U}(q, p), \underline{U}(\tau^*)\}$ , and we immediately have the following lemma:

**Lemma 9.** *When the low-type signals his type, the unique payoff for him is  $\underline{U}_0$ .*

The reason for the above result is that the “intuitive criterion” selects the unique “Riley outcome”.

### 3.2.2. The equilibrium

Next, consider the following situation: given the seller's current belief is her prior, and it is the seller's turn to make an offer. Knowing in the subsequent signaling game the buyer will signal his type, the seller solves the same (SP) as before, except that the right-hand side of (PC) is replaced by  $\delta\underline{U}_0$  now. Solving (SP), the menu could consist of either two self-selection contracts or a single contract depending on the parameter values.

When the menu consists of two self-selection contracts, there are two possibilities: among the four constraints, either (IC) and (PC) are binding, or (IC), (PC) and (PC) are binding. When (IC) and (PC) are binding, solving this problem, we obtain the menu  $\{(q^0, p^1); (\bar{q}^*, \bar{p}^1)\}$ , where  $q^0$  is defined before,  $p^1 = R(q^0, \underline{\theta}) - \delta\underline{U}_0$  and  $\bar{p}^1 = R(\bar{q}^*, \bar{\theta}) - R(q^0, \bar{\theta}) + p^1$ . When (IC), (PC) and (PC) are binding, solving this problem, we obtain the menu  $\{(q^0, p^1); \bar{m}_s^*\}$ , where the low-type's contract is the same as the previous case, but the high-type obtains the complete information contract  $\bar{m}_s^*$ .

The following lemma characterizes the condition for the above two possibilities (see also Jullien, 2000):

**Lemma 10.** *When the menu offered by the seller consists of two self-selection contracts, there are two possible cases. If the following condition holds:*

$$\delta\bar{S}^* \leq (1 + \delta)[R(q^0, \bar{\theta}) - R(q^0, \underline{\theta}) + \delta\underline{U}_0], \quad (\text{C5})$$

*menu  $\{(q^0, p^1); (\bar{q}^*, \bar{p}^1)\}$  is offered; otherwise, menu  $\{(q^0, p^1); \bar{m}_s^*\}$  is offered.*

<sup>17</sup> This case cannot happen in Sen (2000) because of his Assumption 2. See also footnote 16 of Sen (2000). In our model, we do not make such an assumption and we consider all possible parameter values of the two types.

When condition (C5) holds, just like the free signaling case, there are two possible equilibria, and each of them is unique under certain parameter values as specified in Proposition 3.

The simultaneous separating equilibrium (SIE2). In this equilibrium, the seller proposes menu  $\{(q^0, \underline{p}^1); (\bar{q}^*, \bar{p}^1)\}$ , and each type buyer accepts this menu without delay and selects the contract for himself. In this case, the payoff for the seller is  $V_1 = \pi[p^1 - C(q^0)] + (1 - \pi)[\bar{p}^1 - C(\bar{q}^*)]$ .

The sequential separating equilibrium (SEE2). In this equilibrium, the seller proposes  $\bar{m}_s^*$ . The high-type buyer accepts it without delay, but the low-type buyer rejects the offer and makes a counteroffer. The counteroffer is either  $\underline{m}_b^*$  with delay  $\tau^*$  or  $(p_1, q_1)$  without delay, depending on the parameter values as Proposition 2 indicates. The seller accepts this counteroffer without delay. The payoff for the seller is either  $V_{11} = \pi \frac{\delta^{\tau^*+2}}{1+\delta} \underline{S}^* + (1 - \pi) \frac{1}{1+\delta} \bar{S}^*$  when delay is the signal, or  $V_2 = \pi \frac{\delta^2}{1+\delta} \underline{S}^* + (1 - \pi) \frac{1}{1+\delta} \bar{S}^*$  if quantity distortion is the signal.<sup>18</sup> Since  $V_{11} < V_2$ , the seller would prefer the low-type to signal by quantity distortion. However, the low-type will choose the signal in order to maximize his own payoff. We can obtain the following result:

**Proposition 3.** Under conditions (C3) and (C5), there exists  $0 < \pi^1 < 1$  such that if  $\pi \leq \pi^1$ , there exists the unique sequential separating equilibrium (SEE2); if  $\pi > \pi^1$ , there exists the unique simultaneous separating equilibrium (SIE2).

As in the free signaling case before, if the probability of the low-type is sufficiently small, it is better for the seller to exclude the low-type in the first offer. Because by including it, the seller has to pay the high-type buyer information rent and the gain from the low-type cannot compensate for the information rent. On the other hand, if the probability of the low-type is sufficiently large, it is better for the seller to propose the self-selection menu.

Furthermore, we claim that the equilibrium specified above is unique. The intuition is as follows. First, the seller's problem is well-defined. The PC for each type buyer takes his own unique continuation payoff as the outside option, and the ICs ensure there is no pooling contracts. Therefore, the solution to the seller's problem is unique. Depending on the parameter values, the seller proposes either a menu if  $\pi > \pi^1$  or a single contract if  $\pi \leq \pi^1$ . The two type buyers will respond accordingly. All of the players obtain their corresponding unique payoffs. As a consequence, the equilibrium, either SIE2 or SEE2, is unique.

However, when condition (C5) is violated, there is one possible equilibrium: the simultaneous separating equilibrium (SIE3). In this equilibrium, the seller proposes menu  $\{(q^0, p^1); \bar{m}_s^*\}$ , and each type accepts this menu without delay and selects the contract for himself. The payoff for the seller is  $V'_1 = \pi[p^1 - C(q^0)] + (1 - \pi) \frac{1}{1+\delta} \bar{S}^*$ .

**Proposition 4.** Under condition (C3), when condition (C5) is violated, there exists the unique simultaneous separating equilibrium (SIE3).

Since  $\bar{m}_s^*$  has been included in the menu, there is no sequential separating equilibrium in this case. Given the parameter values and the seller's well-defined problem, the two type buyers' payoffs, as well as the seller's payoff, are unique. Therefore, SIE3 is the unique equilibrium.

It is interesting to compare the program (SP) above with the standard non-linear pricing model. The objective functions are exactly the same for both, as well as the ICs. The only difference

between the two problems is the right-hand sides of the PCs. The right-hand sides of PCs for the standard non-linear model are usually the outside options, while for (SP), they are the endogenous type-dependent continuation payoffs for the low-type and the high-type, respectively. When  $\pi$  is sufficiently large, in SIE1, SIE2, and SIE3, the contract bargaining game shrinks to the usual contract design game and it ends without delay. In this sense, we call menus  $\{(q^0, \underline{p}^0); (\bar{q}^*, \bar{p}^0)\}$ ,  $\{(q^0, \underline{p}^1); (\bar{q}^*, \bar{p}^1)\}$  or  $\{(q^0, \underline{p}^1); \bar{m}_s^*\}$  the “bargaining-proof contracts”.

### 3.2.3. Delay or quantity distortion persist as $\Delta \rightarrow 0$

In the previous discussions, the time interval  $\Delta$  is normalized to be one. Now we focus on arbitrarily small value of  $\Delta$  and let  $\Delta \rightarrow 0$ .<sup>19</sup>

Let  $r$  be the common discount rate. Then the common discount factor per period of length  $\Delta$  is  $\delta(\Delta) = \exp(-r\Delta)$ . Note that when  $\Delta \rightarrow 0$ , we should have  $\delta(\Delta) \rightarrow 1$ .<sup>20</sup>

Note that in equilibrium SEE2 when the low-type buyer signals by contract offer  $\underline{m}_b^*$  with delay  $\tau^*$ , the delay will persist as  $\Delta \rightarrow 0$ . Since (IC<sup>1</sup>) is binding when solving the low-type buyer's problem above, we can obtain  $\tau^*$  directly:

$$\delta^{\tau^*} = e^{-r\Delta\tau^*} = \frac{\bar{S}^*}{(1 + \delta)[R(q^*, \bar{\theta}) - C(q^*)] - \delta \underline{S}^*}.$$

Therefore, when  $\Delta \rightarrow 0$ , we obtain the following result:

$$\lim_{\Delta \rightarrow 0} \tau^* = \ln \frac{2[R(q^*, \bar{\theta}) - C(q^*)] - \underline{S}^*}{\bar{S}^*} > 0.$$

In words, when the low-type buyer signals his type by strategic delay,  $\tau^*$  should be long enough so that the high-type will not mimic, which is independent of the time length  $\Delta$ .

Note that in SIE1, SIE2 and SIE3, the quantity distortions are also independent of  $\Delta$ . Therefore, the quantity distortions persist in SIE1, SIE2 and SIE3 as  $\Delta \rightarrow 0$ .

In summary, the above analysis shows that analyzing the contract bargaining game is not as complicated as one might think. It is just the combination of a screening problem and a signaling problem. First, we solve the signaling problem and find the “Riley outcome”, then we use the payoffs of the “Riley outcome” as the endogenous type-dependent continuation payoffs for the PCs in the screening problem. Finally we solve the screening problem to obtain the results.

Usually, one obtains multiple equilibria in sequential bargaining with alternating offers and incomplete information. However, we obtain unique PBE in the contract bargaining game. The reason is because our bargaining contract game is a multidimensional one with strategic delay, and the players can combine those dimensions and use the strategic delay either to signal or to screen credibly, which helps to obtain the unique equilibrium.

## 4. Conclusion

Despite the fact that many real-life contract negotiations involve bargaining, the standard contract theory has not included bargaining in its model. It abstracts away bargaining by assuming that the principal has all of the bargaining power and she makes a take-it-or-leave-it offer. Following Sen (2000) and Admati and Perry (1987), this article introduces alternating offers and strategic delay into the non-linear pricing model. In the free signaling and

<sup>19</sup> See Section 3.2.4 in Muthoo (1999) for reasons why it is important to discuss this case.

<sup>20</sup> See also Section 7 of Admati and Perry (1987).

<sup>18</sup>  $V_2$  is also defined in the free signaling case.

the costly signaling cases, if the initial probability for the low-type is sufficiently small, we obtain the sequential separating equilibrium; otherwise, we achieve the simultaneous separating equilibrium. We prove the existence and the uniqueness of the equilibrium. When time between offers approaches zero, we show that either delay or quantity distortion persist.

The results of this paper are ready to be applied to various contract bargaining environments. For example, the firm and the labor union bargain over labor contracts which specify the wage and the workload; the borrower and the lender bargain over financial contracts which specify the interest rate and the size of the loan; and the insurance company and the customer bargain over insurance contracts which specify the premium and the benefit.

There are also some interesting extensions on our model. For example, following the bargaining literature, the model can be extended to two-sided private information or cases with more than two types. Furthermore, many other bargaining rules, as well as various bargaining power distributions, can be implemented in addition to the one we have studied.

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## Appendix

**Proof of Lemma 1.** Note that the seller will reject any offer which gives her less than  $\frac{\delta}{1+\delta}\bar{S}^*$ . In addition, each type buyer can guarantee himself the equilibrium payoff of no less than his complete information payoff in the free signaling case.

Suppose a buyer proposes a pooling contract that both type buyers are made better off compared to their complete information contracts. The seller will reject the pooling contract. Otherwise, she obtains less than  $\frac{\delta}{1+\delta}\bar{S}^*$ .

Suppose that a buyer proposes a contract  $m$ , which makes the low-type's payoff  $\underline{U}(m) > \frac{1}{1+\delta}\bar{S}^*$  and the high-type's payoff  $\bar{U}(m) < \frac{1}{1+\delta}\bar{S}^*$ . The seller will infer that this buyer is a low-type. We show the seller can profitably deviate by offering  $\bar{m}$  to let  $\underline{U}(\bar{m}) = \delta(\underline{U}) + \xi$  for some  $\xi > 0$ . The low-type will accept this offer. The seller's payoff will be  $\delta[\bar{S}^* - \delta(\underline{U}) - \xi]$ . Since  $\underline{U} > \frac{1}{1+\delta}\bar{S}^*$ , we can get  $\delta[\bar{S}^* - \delta(\underline{U})] > [\bar{S}^* - \underline{U}]$ . Then there exists an  $\xi > 0$  such that the above deviation is profitable for the seller. This proves  $\underline{U} \leq \frac{1}{1+\delta}\bar{S}^*$ . In the case of free signaling, when the low-type buyer makes an offer, he maximizes his payoff by offering his complete information contract  $\underline{m}_b^*$  and obtains the unique payoff  $\underline{U} = \frac{1}{1+\delta}\bar{S}^*$ . The same argument holds for the high-type buyer.  $\square$

**Proof of Proposition 1** (*The Existence of the Equilibrium*). By  $V_1 > V_2$  (and also footnote 14), we obtain

$$\pi > \pi^0 \equiv \max \left\{ \frac{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - R(\underline{q}^0, \underline{\theta})) + \delta(\bar{S}^* - \bar{S}^*)}{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - C(\underline{q}^0)) - \delta(\bar{S}^* + \bar{S}^*)}, 1 - \frac{R'(\underline{q}, \underline{\theta})}{R'(\underline{q}, \bar{\theta})} \right\}.$$

There exists  $0 < \delta < 1$  such that  $0 < \pi^0 < 1$ .

We now begin to prove the existence of the equilibrium. In the following we prove the claim: “if  $\pi^0 < \pi$ , the simultaneous separating equilibrium (SIE1) exists”. By the same token, we can also prove the existence of the sequential separating equilibrium (SEE1).

**Proof.** Suppose  $\pi^0 < \pi$ . Then the following is an equilibrium.

- The seller's beliefs: Suppose that in response to the last offer  $m$  made by the seller, the buyer offers  $\tilde{m}$  after a delay of  $\tau$ . (1) Suppose  $\bar{U}(m) \geq \bar{U}(\bar{m}_b^*)$ . If  $\bar{U}(m) \geq \bar{U}(\tilde{m}, \tau)$ ,  $\underline{U}(\tilde{m}, \tau) > \underline{U}(m)$ , and  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , then  $\pi = 1$ . Otherwise  $\pi = 0$ . (2) Suppose  $\bar{U}(m) < \bar{U}(\bar{m}_b^*)$ . If  $\bar{U}(\bar{m}_b^*, \tau) \geq \bar{U}(\tilde{m}, \tau)$ ,  $\underline{U}(\tilde{m}, \tau) > \underline{U}(m)$ , and  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , then  $\pi = 1$ . Otherwise,  $\pi = 0$ .
- The seller's strategy: The seller offers menu  $\{(q^0, p^0); (\bar{q}^*, \bar{p}^0)\}$  at time zero. If in response to this offer, the buyer makes a counteroffer  $\tilde{m}$  after delay  $\tau$ , and the seller's belief at the node is  $\pi$ :
  - (1) If  $\pi = 0$ , then the seller accepts  $\tilde{m}$  if  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , otherwise it rejects the offer and counteroffers  $\bar{m}_b^*$  without delay.
  - (2) If  $\pi = 1$ , then the seller accepts  $\tilde{m}$  if  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , otherwise it rejects the offer and counteroffers  $\bar{m}_b^*$  without delay.
- The high-type buyer's strategy: At any point, if the seller makes a menu offer including a contract  $m$ , the high-type buyer accepts it and chooses  $m$  without delay if  $\bar{U}(m) \geq \bar{U}(\bar{m}_b^*)$ ; otherwise, it rejects the offer and counteroffers  $\bar{m}_b^*$  without delay.
- The low-type buyer's strategy: At any point, if the seller offers a menu including a contract  $m$ , the low-type buyer accepts it and chooses  $m$  without delay if  $\underline{U}(m) \geq \underline{U}(\underline{m}_b^*)$ ; otherwise, it rejects the offer and counteroffers  $\underline{m}_b^*$  without delay.
- Since  $\pi^0 < \pi$ , as we have shown above, offering  $\{(q^0, p^0); (\bar{q}^*, \bar{p}^0)\}$  is optimal for the seller. The rest of the verification is straightforward. Note that in this equilibrium, the players' strategies are optimal given the beliefs, and the beliefs are updated by using Bayes' rule whenever possible.  $\square$

**Proof of Lemma 4.** First, the seller will reject any pooling offer which gives her payoff no more than  $\frac{\delta}{1+\delta}\bar{S}^*$  because she can guarantee herself the payoff  $\frac{\delta}{1+\delta}\bar{S}^*$  even when  $\pi = 1$ ; with  $\pi < 1$ , she expects to obtain more than  $\frac{\delta}{1+\delta}\bar{S}^*$ . Second, suppose there exists a pooling offer  $m$  such that  $V(m) > \frac{\delta}{1+\delta}\bar{S}^*$  and  $\bar{U}(m) \geq \frac{1}{1+\delta}\bar{S}^*$  and the seller will accept it. But the low-type can deviate profitably by offering  $m'$  such that  $\underline{U}(m') > \underline{U}(m)$ ,  $\bar{U}(m') < \bar{U}(m)$  and  $V(m') \geq V(\bar{m}_b^*)$ . Therefore, no pooling offer can exist in the equilibrium path.  $\square$

**Proof of Lemma 5.** We first show  $\sup(\bar{U}) \leq \frac{1}{1+\delta}\bar{S}^*$  when the high-type makes an offer. Suppose  $\sup(\bar{U}) = \frac{1}{1+\delta}\bar{S}^* + \alpha$  for some  $\alpha > 0$ . This implies that the seller can realize in the respective continuation game a payoff not bigger than  $\frac{\delta}{1+\delta}\bar{S}^* - \alpha$ . We show that the seller can profitably deviate by rejecting and offering a contract  $\bar{m}(\bar{q}^*, \bar{p})$  such that

$$\bar{q}^* - \bar{p} = \delta \left( \frac{1}{1+\delta}\bar{S}^* + \alpha \right) + \alpha \frac{1-\delta^2}{2\delta}.$$

With this contract,  $\bar{U}(\bar{m}) \geq \delta(\frac{1}{1+\delta}\bar{S}^* + \alpha)$ , and the high-type will not reject it. It remains to be shown that this strategy is strictly profitable to the seller. To see this, note:

$$\begin{aligned} \delta[\bar{p} - C(\bar{q}^*)] - \left[ \frac{\delta}{1+\delta}\bar{S}^* - \alpha \right] \\ = \frac{\alpha(1-\delta^2)}{2} > 0. \end{aligned}$$

This proves that  $\sup(\bar{U}) \leq \frac{1}{1+\delta} \bar{S}^*$  when the high-type makes an offer. In the least-cost separating equilibrium, the high-type maximizes his payoff by offering  $\bar{m}_b^*$  without delay and obtains  $\frac{1}{1+\delta} \bar{S}^*$ .  $\square$

**Proof of Lemma 6.** We want to show  $\sup(V) = \frac{\delta}{1+\delta} \bar{S}^*$  when  $\pi = 1$  and when it is the seller's turn to reject or accept the low-type's offer. The proof is similar to that of Lemma 5.  $\square$

**Proof of Lemma 10.** When (IC) and (PC) are binding, the high-type attains its complete information payoff plus the information rent. The information rent is  $R(\underline{q}^1, \bar{\theta}) - R(\underline{q}^1, \underline{\theta}) + \delta \underline{U}_0 - \frac{\delta}{1+\delta} \bar{S}^* > 0$ .  $\square$

**Proof of Proposition 3 (Existence).** By comparing  $V_1$  and  $V_2$ , (and also considering the case in footnote 14), we define:

$$\pi^{1'} \equiv \max \left\{ \frac{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - R(\underline{q}^0, \underline{\theta}) + \delta \underline{U}_1) - \delta \bar{S}^*}{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - C(\underline{q}^0)) - \delta(\delta \bar{S}^* + \bar{S}^*)}, 1 - \frac{R'(\underline{q}, \underline{\theta})}{R'(\underline{q}, \bar{\theta})} \right\}.$$

By comparing  $V_1$  and  $V_{II}$ , (and also considering the case in footnote 14), we define

$$\pi^{1''} \equiv \max \left\{ \frac{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - R(\underline{q}^0, \underline{\theta})) + \delta(\delta \tau^* \bar{S}^* - \bar{S}^*)}{(1+\delta)(R(\underline{q}^0, \bar{\theta}) - C(\underline{q}^0)) - \delta(\delta \tau^{*+1} \bar{S}^* + \bar{S}^*)}, 1 - \frac{R'(\underline{q}, \underline{\theta})}{R'(\underline{q}, \bar{\theta})} \right\}.$$

Let  $\pi^1 = \pi^{1'}$  when (C4) holds; while let  $\pi^1 = \pi^{1''}$  otherwise.

Now we begin to prove the existence of the equilibrium. In the following we prove the claim: “if  $\pi \leq \pi^1$ , there exists the sequential separating equilibrium (SEE2)”. By the same token, we can prove the existence of the simultaneous separating equilibrium (SIE2).  $\square$

**Proof.** Proof. Suppose that  $0 < \pi \leq \pi^1$ , then the following is an equilibrium.

- The seller's beliefs: Suppose in response to the last offer of  $m$  by the seller, the buyer offers  $\tilde{m}$  after a delay of  $\tau$ . (1) Suppose  $\bar{U}(m) \geq \bar{U}(\bar{m}_s^*)$ . If  $\bar{U}(m) \geq \bar{U}(\tilde{m}, \tau)$ ,  $\underline{U}(\tilde{m}, \tau) > \underline{U}(m)$ , and  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , then  $\pi = 1$ . Otherwise  $\pi = 0$ . (2) Suppose  $\bar{U}(m) < \bar{U}(\bar{m}_s^*)$ . If  $\bar{U}(\bar{m}_b^*, \tau) \geq \bar{U}(\tilde{m}, \tau)$ ,  $\underline{U}(\tilde{m}, \tau) > \underline{U}(m)$ , and  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , then  $\pi = 1$ . Otherwise,  $\pi = 0$ .
- The seller's strategy: The seller offers  $\bar{m}_s^*$  at time zero. If in response to an offer  $m$  by the seller, the buyer makes a counteroffer  $\tilde{m}$  after delay  $\tau$ , and the seller's belief at this node is that the buyer is a low-type with probability  $\pi$ , then:
  - (1) If  $\pi = 0$ , then the seller accepts  $\tilde{m}$  if  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , otherwise it rejects the offer without delay and counteroffers  $\bar{m}_s^*$ .
  - (2) If  $\pi = 1$ , then the seller accepts  $\tilde{m}$  if  $V(\tilde{m}, \tau) \geq V(\bar{m}_b^*, \tau)$ , otherwise it rejects the offer without delay and counteroffers  $\bar{m}_s^*$ .
- The high-type buyer's strategy: At any point, if the seller makes an offer  $m$ , the high-type buyer accepts it without delay if  $\bar{U}(m) \geq \bar{U}(\bar{m}_s^*)$ ; otherwise it rejects the offer without delay and counteroffers  $\bar{m}_b^*$ .
- The low-type buyer's strategy: At any point, if the seller offers  $m$ , the low-type buyer accepts it without delay if  $\underline{U}(m) \geq \delta \underline{U}_0$ ; if  $\underline{U}(m) < \delta \underline{U}_0$ , it rejects the offer and counteroffers either  $(\underline{p}_1, \underline{q}_1)$  when (C4) holds or  $\bar{m}_b^*$  with delay  $\tau^*$  otherwise.

- To verify the equilibrium first note that the seller would not offer  $m$  such that  $\bar{U}(m) < \bar{U}(\bar{m}_s^*)$ , since given the buyer's strategy this leads to a lower payoff than offering  $\bar{m}_s^*$ . Since  $0 < \pi \leq \pi^1$ , as we have discussed above, offering  $\bar{m}_s^*$  is optimal for the seller. The rest of the verification is straightforward. Note that the beliefs above satisfies (A1) and are updated by using Bayes' rule whenever possible, and the strategies are optimal given the beliefs.  $\square$

**Proof of Proposition 3 (Uniqueness).** In the following, we show when  $\pi^1 < \pi$ ,  $V_1$  is the unique payoff for the seller and the only equilibrium path is SIE2.

First, as mentioned before, the seller's problem is well-defined. The ICs ensure there is no pooling contract, and the corresponding PC takes the unique payoff for each type buyer as his outside option. Second, as we have shown, when  $\pi > \pi^1$ , the menu  $\{(q^1, p^1); (\bar{q}^*, \bar{p}^1)\}$  is the unique solution to the seller's problem. As a consequence,  $V_1$  is the unique payoff for the seller. Also in SIE2, the high-type has a higher payoff than when he proposes, and the low-type gets his unique payoff as Lemma 10 shows. Both types will accept the offer without delay. Thus, the SIE2 is the only candidate for an equilibrium path. The existence of an equilibrium that supports this path has been established above.

The above proof shows that SIE2 is unique. By the same token, we can prove that SEE2 is unique.  $\square$

**Proof of Proposition 4.** The proofs for the existence and the uniqueness are similar to those for Proposition 3. Therefore, we will skip the details.  $\square$

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